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March 26, 1965
Revision of Int. Fed. Spec
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June 17, 1964

INTERIM FEDERAL SPECIFICATION

GLOVES, SURGEON'

This Interim Federal Specification was developed by the Federal Supply Service, General Services Administration, Washington, D. C., 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address above. The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification ZZ-G-421a, dated April 17, 1940.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers natural latex rubber and natural rubber cement gloves suitable for surgical and obstetrical procedures.

1.2 Classification. Gloves covered by this specification shall be of the following grades and types, as specified in the invitation for bids (6.2).

Grade A. Natural rubber latex.
Type I. General surgical gloves.
Type II. Obstetrical gloves

Grade B. Natural rubber cement.
Type I. General surgical gloves.
Type II. Obstetrical gloves.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner
PPP-B-636 - Box, Fiberboard
PPP-B-676 - Boxes, Set-up, Paperboard

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 601 - Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

FSC 6515

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case, & Sheet, Overwarp, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publication: Of the issue in effect on date of invitation for bids, forms a part of this specification.

United States Pharmacopeial Convention, Inc.
Pharmacopeia of the United States of America.
(Application for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania).

3. REQUIREMENTS

3.1 Material

3.1.1 Grade A. Grade A gloves shall be fabricated of natural rubber latex.

3.1.2 Grade B. Grade B gloves shall be fabricated of natural rubber cement(Brown Milled).

3.2 Design.

3.2.1 Edge of cuff. The edge of the cuff shall be rolled outward and backward to form a self-binding roll, or left flat, or reinforced along the edge with a band, or shall be a combination of both roll and band.

3.2.2 Fingers. The gloves shall be four fingers and opposed thumb style (see fig. 1) made on standard commercial surgical glove forms. A pair of gloves shall consist of one right-hand glove and one left-hand glove. The base of the thumb on the palm side shall be continued to the glove palm in such a manner that the thumb-crotch area shall show no noticeable creases. Fingers shall be curved (6.2) if specified in the invitation for bid.

3.3 Finish. Finish of the gloves shall be smooth, or shall be roughened (6.2) if specified in the invitation for bid.

3.4 Sizes. Gloves procured under this specification shall be of the sizes listed in Table I, and as specified in the invitation for bids (6.2).

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TABLE I. Length and palm circumference.

Type I (General Surgical Gloves)

Size (Circumference of palm[1] in inches)	Length (Inches) Min. Grade A	Grade B
6	10	10-3/4
6-1/2	10	10-3/4
7	10	10-3/4
7-1/2	10-1/2	10-3/4
8	10-1/2	10-3/4
8-1/2	10-1/2	10-3/4
9	10-1/2	10-3/4
9-1/2	10-1/2	10-3/4
10	10-1/2	10-3/4
11	10-1/2	10-3/4

[1]A tolerance of -1/8 or ∱ 1/4 inch in circumference will be permitted.

TABLE I. Length and palm circumference.

Type II (Obstetrical Gloves)

(Circumference of palm[1] in inches)	Length (Inches) Min. Grade A	Grade B
7	16, 18, 21	18
7-1/2	16, 18, 21	18
8	16, 18, 21	18
8-1/2	16, 18, 21	18
9	16, 18, 21	18

[1]A tolerance of -1/8 or ∱ 1/4 inch in circumference will be permitted.

3.5 Dimensions.

3.5.1 Length. Length shall be as specified in Table I, when measured in accordance with 4.4.5.1. Length of Grade A, Type II gloves (obstetrical) shall be as specified in the invitation for bids (6.2).

3.5.2 Circumference. Palm circumference shall be as specified in Table I, when measured in accordance with 4.4.5.2.

3.5.3 Thickness. Thickness shall be as specified in Table II, when tested in accordance with 4.5, except the thickness at any place on the glove shall not be less than 0.004 inch.visible thin spots (4.4) shall be measured.

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TABLE II. Thickness dimensions. All dimensions are in inches

Type I (General Surgical Gloves)

Grade	Finger Tip		Palm		Crotch		Wrist	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
A	0.008	0.015	0.008	0.013	0.007	0.013	0.006	0.013
B	0.006	0.009	0.007	0.010	0.006	0.009	0.007	0.010

Type II (Obstetrical Gloves)

A	0.008	0.015	0.008	0.013	0.008	0.013	0.007	0.013
B	0.008	0.011	0.009	0.011	0.009	0.011	0.009	0.011

3.6 Marking. Gloves shall be stamped with the manufacturer's name or trade mark, and glove size. Markings shall remain legible when sterilized in accordance with 4.6.2.

3.7 Color. Grade A gloves shall be natural or brown. Grade B gloves shall be brown.

3.8 Color Coding. When specified in the invitation for bids (6.2), the gloves shall be color coded as follows, 6-1/2 shall be blue; 7 shall be red; 7-1/2 shall be black; 8 shall be green; all others shall be color coded yellow. The color coding shall be an integral part of the glove, and shall remain legible when sterilized in accordance with 4.6.2. The color coding shall be either the flat band described in 3.2.1, or shall be a seperate band, or color coded size numbers.

3.9 Dusting powder. Each pair of Grade A gloves shall be dusted with absorbable dusting powder conformint to the requirements of the United States Pharmacopeia. Grade B gloves may be dusted.

3.10 Mechanical properties. Mechanical properties shall be as expressed in Table III, when tested in conformance with 4.5. Accelerated aging and sterilization shall be conducted as described in 4.6.1 and 4.6.2.

3.11 Workmanship. The gloves shall be free from defects listed in Table V.

TABLE III. Mechanical Properties

Grade	Column 1 Unaged Unsterilized Specimens			Column 2 Aged Specimens			Column 3 Sterilized Specimens	
	Modulus (Tensile Stress) at 500 percent Elongation (Max.) (lb per sq. in.)	Tensile Strength (Min.) (lb per sq. in.)	Ultimate Elongation (Min.) Percent	Duration of aging (Hours)	Tensile Strength (Min.) (lb per sq. in.)	Ultimate Elongation (Min.) percent	No. of Sterlizations	Tensile Strength (Min.) lb per sq. in.
A	800	4000	800	96	3000	800	5	4000
							10	3500
B	800	3000	700	96	1500	700	5	2500
							10	2000

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and test shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern. When available, certificates of quality, supplied by the manufacturer of the component of material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.3 Sampling.

4.3.1 Inspection lot. The inspection lot shall be determined in accordance with MIL-STD-105. The inspection lot shall consist of all the gloves of the same grade and type offered for delivery at the same time.

4.3.2 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection level and Acceptable Quality Level (AQL) shall be as indicated in Table IV. Unit for examination shall be a glove, but selection shall be by pairs to insure examination of both right-hand and left-hand gloves. Defects shall be as listed in Table V.

TABLE IV. Sampling for examination

Defect Type	Inspection Level	AQL (Percent Defective)
Visual Examination		
Major	I	1.0
Minor	I	4.0
Dimensional Examination (including thickness)	S-3	2.5

4.3.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Properties to be tested and the applicable test method shall be as stated in Table VII. The unit for testing shall be a pair of gloves. A MAXIMUM of 2 units should be tested for lots up to 100 dozen. SAMPLING PLAN S-1 shall be used for lots of 100 dozen and over, where possible, the same sample shall be used for the determination of two or more test characteristics (Table VII). Failure of any sample or specimen to comply with any test requirement, shall be cause for rejection of the lot represented (4.3.1).

4.4 Examination.

4.4.1 Design. Gloves shall be examined for design to determine conformance with 3.2.1 and 3.2.2.

4.4.2 Workmanship. Gloves shall be examined for defects listed in Table V. Examination shall include inflation of the gloves in sufficient amount to observe for defects listed in Table V.

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4.4.3 Sizes. Gloves shall be examined to determine that they are the sizes specified in 3.4, and are as specified in the invitation for bids (6.2).

4.4.4 Type. Gloves shall be examined to determine that they are of the type specified in the invitation for bids (6.2).

4.4.5 Dimensions.

4.4.5.1 Length. Length of the glove shall be measured from the tip of the middle finger to the edge of the cuff as described in Method 2411 of Federal Test Method Standard 601.

4.4.5.2 Circumference. The outer circumference of the glove shall be measured at a point between the base of the index finger and the thumb by the method described in Method 2331 of Federal Test Method Standard 601.

TABLE V. Visual and Dimensional Defects classified according to Categories

Visual Examination		Dimensional Examination
Major Defects	Minor Defects	
1. Pinholes	1. Blisters(small clusters)	1. Length(shorter than specified in Table I; Type II, Grade A Gloves not length specified in 6.2
2. Particles(Large; imbedded; ingrained)	2. Marking(incomplete; missing; illegible; incorrect; not legible after sterilized in accordance with 4.6.2)	2. Palm circumference exceeds tolerances specified.
3. Tears; Large holes	3. Color coding(missing; incorrect; not as specified in 6.2; loosely attached; indistinct)	
4. Design defects	4. Dusting powder missing	
5. Size, not as specified (6.2)		
6. Grade and/or type not as specified (6.2)		
7. Blisters(Large; open)		
8. Visible thin spots(3.5.3)		

4.4.6 Marking. Gloves shall be examined to determine that they are marked in accordance with 3.6.

4.4.7 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in Section 5 of this specification. Defects shall be scored in accordance with Table VI. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-3 and the AQL shall be 4.0 per hundred units.

TABLE VI. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadquate stapling. Bulging or disotortion of container.
Packaging and Packing	Case liners not sealed. Contents more or less than specified.

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4.5 Tests. Tests, mechanical properties, and thickness shall be conducted in accordance with Fed. Test Method Std. 601. Test methods and specimens shall be as listed in Table VII. In measuring thickness, measurements shall include visible thin spots (4.4, Table V), places listed in Table II, and as shown in figure 1, and all four fingers and thumb.

TABLE VII. (Federal Test Method Standard 601)

Property	Test Method Number	Specimen
Tensile Stress	4131	4111 (Dumbell; Die III)
Tensile Strength	4111	4111 (Dumbell; Die III)
Ultimate Elongation	4121	4111 (Dumbell; Die III)
Thickness	2011	20001

4.6 Accelerated aging and sterilization procedures.

4.6.1 Accelerated aging. Samples (4.3.3) and specimens (4.5) shall be protected against exposure to strong light prior to aging in light-proof containers, except while being inspected, prepared, or measured. Specimens shall be aged in accordance with Test Method 7111 of Fed. Test Method Std. 601, except that duration of aging shall be as specified in Table I. Specimens shall then be tested as specified in 4.5.

4.6.2 Sterilization. Samples (4.3.3) shall be protected against exposure to strong light as specified in 4.6.1. Samples shall be sterilized in accordance with test Method 7411 of Fed. Test Method Standard 601, except the units for sterilizations may be gloves (4.3.3) or dumbell strips; limitation of air per 1000 volumes of steam shall not apply, but rather that air be flushed from the sterilizer between sterilizations. At the end of sterilizations the units shall be removed and tested in accordance with 4.5 to determine compliance with 3.11.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and application of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A, B, or C as specified in the invitation for bids (6.2).

5.1.1 Unit of issue. One (1) pair of gloves, of the grade, type and size (6.2), constitutes one unit of issue. Each pair of gloves of the same grade, type and size shall be tissue wrapped.

5.1.2 Packaging quantities. The number of units of issue indicated in Table VIII shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

TABLE VIII. Packaging Quantities

Unit package	Intermediate package	Exterior container
12 Pr	72 Pr	432 Pr

5.1.2.1 Packing quantity variation. If the required number of units to be shipped is less than the number of units specified to be packed in an exterior container, such units shall be packed in an exterior container of suitable size conforming to those specified in 5.2 as applicable.

5.1.3 Level A.

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5.1.3.1 Unit packaging. Gloves, of one grade, type and size, in the quantity shown in Table VIII, shall be unit packaged in a box conforming to PPP-B-566, or PPP-B-676. Each box shall be adequately secured to prevent accidental opening.

5.1.3.2 Intermediate packaging. Gloves of one grade, type, and size in the quantity shown in Table VIII, shall be intermediate packaged in a box conforming to PPP-B-566, or PPP-B-676. Each box shall be adequately secured to prevent accidental opening.

5.1.4 Level B.

5.1.4.1 Unit packaging. The gloves shall be unit packaged as specified in 5.1.3.1.

5.1.4.2 Intermediate packaging. The gloves shall be intermediate packaged as specified in 5.1.3.2.

5.1.5 Level C. The gloves shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be Level A, B, or C as specified in the invitation for bids (6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. The exterior container shall conform to PPP-B-621, Class 2, or to PPP-B-636, V3c.

5.2.1.2 Case liner. Each container shall be lined with a case liner conforming to MIL-L-10547. Closure and sealing shall conform to the appendix thereto. Case liners may be omitted from fiberboard containers conforming to PPP-B-636, when the containers are waterproof sealed with tape in accordance with the appendix to the container specification.

5.2.1.3 Strapping. Strapping shall be in accordance with the applicable container specification.

5.2.2 Level B.

5.2.2.1 Exterior container. The exterior container shall conform to PPP-B-621, Class 1, or to PPP-B-636, Type CF, Class domestic. Closure shall be in accordance with the appendix to the applicable container specification.

5.2.3 Level C. The gloves shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Grade.

6.1.1.1 Grade A gloves. Grade A gloves are intended for usage where the greatest stress might be encountered, and where the greatest mechanical strengths are desired.

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6.1.12 Grade B Gloves. Grade B gloves are intended for use where lesser mechanical strengths are required, and where a softer glove is desired.

6.1.2 Type.

6.1.2.1 Type I. Type I gloves are intended for use in general surgical procedures.

6.1.2.2 Type II. Type II gloves are intended for use in general obstetrical procedures.

6.2 Ordering data. Purchaser should exercise any options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Grade and type of gloves (see 1.2).

(3) Roughened finish, is required (see 3.3).

(4) Size (see 3.4).

(5) Color coding (see 3.8).

(6) Length of Grade A, Type II gloves (see 3.5.1).

(7) Level of packaging, packing, and marking required (see 5.1, 5.2, 5.3.1, and 5.3.2)

(8) Curved fingers, if required (see 3.2.2).

6.3 Transportation Description. Transportation description and minimum weights applicable to this commodity are:

Rail:
Gloves, not otherwise indexed by name.
Carload minimum weight 20,000 pounds, subject to Rule 34,
Uniform Freight Classification.

Motor:
Gloves, not otherwise indexed by name.
Truckload minimum weight 12,000 pounds, subject to Rule 115,
National Motor Freight Classification.

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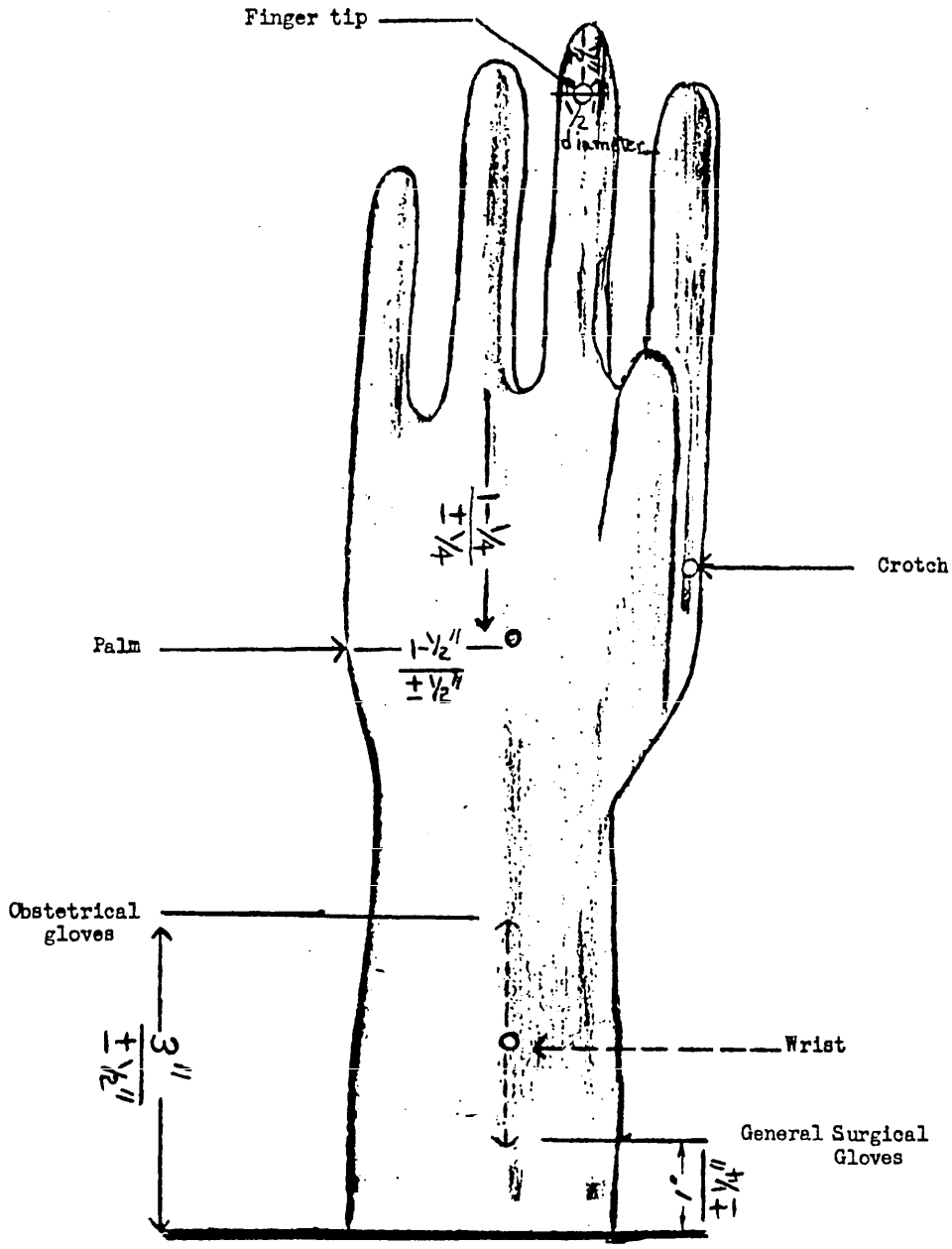


Figure 1. - Thickness measurement

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